FMA4200 Financial Data Analysis: Midterm Exam (Spring, 2025)

Answer the questions in the answer book.

Question:	1	2	3	4	5	6	7	Total
Points:	5	15	15	20	20	10	15	100
Score:								

Name: _____ Student ID: _____ Time limit: 2 hours

Note: $\Phi^{-1}(0.95) = 1.645$ and $\Phi^{-1}(0.975) = 1.96$ where Φ is the cumulative distribution function of the standard normal distribution; i.i.d. stands for independent and identically distributed.

1. True or False. No justification is needed.
   (a) (1 point) The SPX returns are empirically normally distributed.
   (b) (1 point) The distribution of SPX returns is left-skewed and fat-tailed.
   (c) (1 point) The student's t-distribution is more fat-tailed than the normal inverse Gaussian distribution.
   (d) (1 point) An i.i.d. process is always stationary.
   (e) (1 point) An ARMA$(1, 1)$ process is always invertible.

   Total for Question 1: 5

2. Answer the following short questions briefly.
   (a) (3 points) Suppose we have a i.i.d. sample of 1000 stock returns: $x_1, x_2, \ldots, x_{1000}$. How can we estimate the kurtosis of the stock returns?
   (b) (3 points) Suppose we have a i.i.d. sample of 1000 stock returns: $x_1, x_2, \ldots, x_{1000}$. Suppose we have computed the sample mean $\bar{x} = 0.0017$ and the sample variance $s^2 = 0.0001$. Can we conclude that the expected stock return is positive at the 5% significance level?
   (c) (3 points) Continued from the previous question. Can we conclude that the expected stock return is not zero at the 5% significance level?
   (d) (3 points) Suppose we have a i.i.d. sample of 1000 stock returns: $x_1, x_2, \ldots, x_{1000}$. What kinds of tests can we use to check if the stock returns are normally distributed? (Hint: propose at least two tests.)
   (e) (3 points) Suppose we observe a time series. What kind of test can we use to check if the autocorrelations are significantly different from zero?

   Total for Question 2: 15

3. Consider an AR(1) model as follows:
$$X_t = bX_{t-1} + \varepsilon_t,$$
   where $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ and ε_t is i.i.d.
   (a) (5 points) When is X_t stationary?
   (b) (5 points) When is X_t invertible?
   (c) (5 points) Calculate $\mathbb{E}[X_t]$, $\mathrm{Var}[X_t]$, $\mathrm{Cov}(X_t, X_{t-k})$ and $\mathrm{Corr}(X_t, X_{t-k})$ for all $k = 1, 2, \ldots$.

   Total for Question 3: 15

4. Consider the MA(1) model:
$$X_t = \varepsilon_t + \theta\varepsilon_{t-1},$$
   where $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ and ε_t is i.i.d.

(a) (5 points) When is X_t invertible?

(b) (5 points) When is X_t stationary?

(c) (5 points) Calculate $\mathbb{E}[X_t]$, $\mathrm{Var}[X_t]$, $\mathrm{Cov}(X_t, X_{t-k})$ and $\mathrm{Corr}(X_t, X_{t-k})$ for all $k = 1, 2, \ldots$.

(d) (5 points) What is the best prediction of X_{t+1} given the information up to time t? And what is the variance of the prediction error?

Total for Question 4: 20

5. Consider the following ARMA$(1,1)$ model:

$$X_t = bX_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1},$$

where $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ and ε_t is i.i.d.

(a) (5 points) When is X_t invertible?

(b) (5 points) When is X_t stationary?

(c) (10 points) Calculate $\mathbb{E}[X_t]$, $\mathrm{Var}[X_t]$, $\mathrm{Cov}(X_t, X_{t-k})$ and $\mathrm{Corr}(X_t, X_{t-k})$ for all $k = 1, 2, \ldots$.

Total for Question 5: 20

6. Consider the following ARCH(1) model:

$$\begin{cases} X_t = \sigma_t Z_t, \\ \sigma_t^2 = \omega + \alpha X_{t-1}^2, \end{cases}$$

where $Z_t \sim \mathcal{N}(0,1)$ and Z_t is i.i.d.

(a) (5 points) When is X_t stationary?

(b) (5 points) Calculate $\mathbb{E}[X_t]$ and $\mathrm{Var}[X_t]$.

Total for Question 6: 10

7. Consider a European call option on a non-dividend paying stock with strike price K and maturity T. The payoff of the option is given by

$$\max(S_T - K, 0),$$

where S_T is the stock price at time T. We assume that under the physical probability measure $\mathbb{P}$, the stock price follows:

$$S_T = S_0 e^{\mu T + \sigma\sqrt{T}Z},$$

where Z is an exponential random variable with mean 1, i.e., its density function is given by

$$f_Z(z) = e^{-z} \quad \text{for } z \geq 0.$$

The risk-free interest rate is r.

(a) (5 points) Consider a pricing kernel

$$M_T = e^{\alpha + \beta Z},$$

where α and β are constants, and let $\frac{d\mathbb{Q}}{d\mathbb{P}} = M_T$. Determine the values of α and β such that $\mathbb{Q}$ is a risk-neutral probability measure, i.e.,

$$\mathbb{E}_{\mathbb{Q}}[S_T] = S_0 e^{rT}, \ \mathbb{E}_{\mathbb{P}}[M_T] = 1.$$

(b) (10 points) What is the distribution of $\ln S_T$ under the risk-neutral probability measure $\mathbb{Q}$?

(c) (10 points (bonus)) Derive the closed-form expression for the price of the call option and propose a method to estimate the parameters given that we observe the market price of the options with different strike prices $K = K_1, K_2, \ldots, K_n$.

Total for Question 7: 15